

AutiVity – Step-by-Step Activity Expansion Roadmap

Actionable guide prioritized from easiest quick-wins to high-impact routines. Use this checklist as we build each module one by one.

1

START HERE: Pick 'n Choose (Object Identification)

EASIEST QUICK WIN

Target: Add +20 items (Expand from 10 → 30 items).

Why it's first: Requires *no new game logic*. The engine automatically creates 3-option multiple choice grids and text-to-speech audio prompts from simple data entries.

Primary File: `activities/pick-n-choose/data/object-identification.ts`

Category	Items to Add	Asset Key Pattern
Animals (+8)	Dog, Cat, Bird, Fish, Cow, Duck, Lion, Elephant	<code>dog_icon</code> , <code>cat_icon</code> , etc.
Vehicles (+5)	Car, Bus, Bicycle, Airplane, Train	<code>car_icon</code> , <code>bus_icon</code> , etc.
Body Parts (+4)	Eyes, Ears, Hands, Feet	<code>eyes_icon</code> , <code>ears_icon</code> , etc.
Emotions (+3)	Happy, Sad, Angry	<code>happy_face</code> , <code>sad_face</code> , etc.

2

Turn-Taking Cooperative Levels

ZERO IMAGE ASSETS NEEDED

Target: Add +7 cooperative path levels (Expand from 3 → 10 levels).

Why it's second: Requires *no graphic design or illustration files*. Levels are created purely by writing an array of `{ x, y }` coordinate waypoints.

Primary File: `activities/turn-taking/data/levels.ts`

Level #	Name & Path Concept	Difficulty	Target Skill
Level 4	S-Curve (Double wave track)	Difficulty 2	Smooth directional change
Level 5	Arch Bridge (High parabolic arc)	Difficulty 2	Vertical lift & descent
Level 6	Double Zigzag (Rapid 4-point corners)	Difficulty 3	Corner precision & pacing
Level 7	Triangle Track (3-corner relay)	Difficulty 3	Angular turn-taking handover
Level 8-10	Loop-de-Loop, Spiral, and W-Track	Difficulty 3-4	Advanced cooperative stamina

3

Drag & Drop Matching

STRAIGHTFORWARD EXPANSION

Target: Add +15 items (Expand from 35 → 50 items).

Files: `activities/drag-drop/data/matching-fruits.ts` , `matching-colors.ts` , or new subcategory pools.

- **Shape Matching (+6):** Match Circle, Square, Triangle, Star, Rectangle, Heart to their outline baskets.
- **Animal to Habitat/Food (+6):** Dog → Bone, Bird → Nest, Fish → Water, Monkey → Banana.
- **Size Sorting (+3):** Big vs. Small object classification.

4 Sequencing Daily Living (ADL) Routines

CRITICAL EDUCATIONAL NEED

Target: Add +8 routines (Expand from 4 → 12 routines).

Requirement: Needs 3 to 4 sequential step images per routine. Map them in `activities/sequencing/utls/assetDictionary.ts`.

Routine Name	Steps Count	Step Sequence Outline
Getting Dressed	3 steps	1. Put on shirt → 2. Put on pants → 3. Put on jacket
Putting on Shoes	3 steps	1. Put on socks → 2. Put feet in shoes → 3. Fasten/Tie shoes
Packing School Bag	3 steps	1. Open backpack → 2. Put books inside → 3. Zip backpack closed
Making the Bed	3 steps	1. Straighten bedsheet → 2. Pull up blanket → 3. Place pillow
Taking a Bath	4 steps	1. Turn on water → 2. Apply soap → 3. Rinse body → 4. Dry with towel
Eating Lunch	3 steps	1. Open lunchbox → 2. Eat food → 3. Drink water & clean up
Cleaning Up Toys	3 steps	1. Pick up scattered toys → 2. Place in bin → 3. Close lid
Crossing the Street	3 steps	1. Stop at sidewalk → 2. Look left & right → 3. Walk safely across

5 Bubble Pop Modes

GAMEPLAY ENHANCEMENT

Target: Add +3 new pop modes (Expand from 2 → 5 modes).

Primary Files: `activities/bubble-pop/utls/bubble-spawner.ts` and `bubble-assets.ts`.

- **Number Pop:** Pop numbers in sequence (1 → 2 → 3).
- **Alphabet Pop:** Pop target letter bubbles.
- **Shape Pop:** Pop only circle bubbles while avoiding square bubbles.